

MissionWeave: A Bounded Organizational Digital Twin for Mission-Readiness Decisions

Topic: DLA26BZ03-NV011, Digital Twin of the Organization for Enhanced Mission Readiness

Program: DoW SBIR 2026 Release 3, Phase I

Applicant: Robert Ashworth d/b/a LumenCore

Principal Investigator: Robert Ashworth

Period of Performance: Six months

Proposed Price: \$100,000, Firm-Fixed-Price

Technical Readiness Objective: Bounded proof of concept within the topic's TRL/MRL 3-6 range

1. Identification and Significance of the Problem or Opportunity

1.1 Mission problem

DLA must make workforce and process decisions under changing demand, skill availability, system degradation, supply disruption, and increasing use of artificial intelligence. Static organization charts, periodic reports, and unlinked spreadsheets can describe a recent state, but they do not reliably answer time-dependent questions: Which handoff is constraining a critical case? What happens when a specialized role is absent during a surge? Does automation remove work or move the bottleneck downstream? Which recommendation remains useful when an input is late, missing, or wrong?

The opportunity is not merely to visualize an organization. It is to create a disciplined, testable representation that connects mission work to process stages, roles, skills, systems, deadlines, constraints, and human decisions; replays an approved current-state baseline; generates clearly labeled scenarios; and compares interventions under identical conditions. The result must expose uncertainty and failure regions rather than produce a single opaque productivity score.

Full digital twins of organizations present an important scientific limitation. Organizations contain agency, conflict, learning, hidden interdependencies, and emergent behavior that are not captured by a process diagram. MissionWeave therefore does not claim a complete causal twin of DLA. Phase I intentionally narrows the proof of concept to one bounded process and treats the twin as a decision-support model whose assumptions, scope, and evidence can be inspected and revised.

1.2 Proposed innovation

MissionWeave is an event-driven, process-bounded organizational digital twin organized around five linked layers:

- A **mission-to-work graph** represents cases, stages, role and skill requirements, systems, service windows, decision authorities, and substitution constraints.
- A **source registry** records origin, owner, permission basis, schema, time range, exclusion rules, quality status, hash, and allowed use for every dataset or assumption set.
- A **scenario engine** replays nominal, surge, absence, degraded-system, and human-machine teaming conditions while visibly separating observed, approved-assumption, public, and synthetic records.
- A **constrained intervention engine** compares current-state, fixed-role, cross-trained, routing, staffing, automation, and stop-doing options under the same frozen scenarios.
- An **evidence layer** binds source, configuration, software version, metrics, negative cases, and outputs into reproducibility receipts and operator-readable intervention cards.

MissionWeave's technical distinction is the integration of bounded modeling, frozen comparator evaluation, human decision authority, and claim-level provenance. A recommendation is not emitted as an instruction. It is delivered with its objective, affected process and roles, assumptions, expected

range, uncertainty, guardrails, known failure modes, rollback condition, and a human approval field. When the evidence contract is not met, the system must abstain.

1.3 Bounded Phase I process

The initial representative process is **Critical Supply Exception Triage and Disposition**. The start event is a readiness-impacting supply exception or request entering a queue. The end event is a human-approved disposition such as expedite, substitute, defer, request more information, route to another authority, or close. This is an unclassified DLA-relevant process assumption selected for proposal feasibility; it is not represented as a current DLA workflow, DLA-provided data, or DLA approval. During kickoff, a DLA Component may confirm this process or select another bounded process with comparable data and decision structure.

Minimum events include a pseudonymous case identifier, arrival time, case family, criticality, stage start/end, due time, required role/skill, system state, absence state, disposition, and data-quality status. Phase I excludes employment decisions, automated personnel evaluation, contract-award decisions, protected-trait inference, classified logistics data, and operational-command decisions.

1.4 Phase I scientific questions

1. Can a bounded event-driven twin reproduce approved process aggregates within a tolerance frozen before policy selection?
2. Can a frozen intervention policy improve one or more primary outcomes against declared comparators without degrading approved guardrails, or correctly abstain where capacity is inadequate?
3. Are recommendations stable under missing, stale, duplicated, and perturbed inputs?
4. Can an authorized reviewer reproduce each reported result from source, configuration, software, and output hashes?
5. Can a bottom-up business case translate measured process changes into bounded ROI ranges without claiming realized savings?

1.5 Significance to DLA

The Phase I product is designed to answer the topic's immediate requirements: identify and rank high-impact use cases; define necessary sources, acquisition and synthetic-data plans, and success criteria; and produce a business case and MVP roadmap. The bounded approach gives DLA a way to distinguish a useful digital process twin from a broad but unfalsifiable organizational model. It also creates a practical alignment point for DLA J1, J3, J6, or J7: a Component can replace generated assumptions with authorized process definitions and data without changing the evaluation contract.

The topic's 10x productivity objective is treated as a strategic research question, not a Phase I promise. MissionWeave will test measurable process interventions, preserve negative results, identify capacity-breakdown regions, and report what additional change would be required for larger gains. No DLA productivity, readiness, savings, sponsor relationship, customer use, or operational deployment is claimed.

2. Phase I Technical Objectives

2.1 Objectives

Objective 1 - Rank high-impact use cases. Evaluate at least three candidates, including critical supply exception disposition, surge role coverage, and human-machine task allocation, using a frozen rubric for mission relevance, decision value, data feasibility, security/privacy burden, implementation complexity, and transition potential. Select one bounded proof-of-concept process with a documented rationale.

Objective 2 - Define the data and control contract. Produce a source inventory, acquisition path, synthetic-data generation plan, event schema, observed/synthetic labeling rules, permission and retention fields, missing/stale-data treatment, security boundary, and success criteria. Prefer

Government-sanitized data or approved assumptions, then written-permission partner data, then public or synthetic evidence. Fully generated evidence remains feasibility-only.

Objective 3 - Build the bounded proof of concept. Implement the process graph, baseline replay, scenario generator, comparator policies, constrained intervention logic, human-approval gate, intervention cards, and evidence receipts for the selected process.

Objective 4 - Conduct a frozen feasibility evaluation. Freeze the process boundary, development/validation split, comparators, metrics, perturbations, and promotion rules before the confirmatory run. Evaluate nominal, surge, targeted-absence, degraded-system, and combined-stress conditions. Report paired outcomes, uncertainty intervals, negative seeds, absolute performance, and abstention behavior.

Objective 5 - Produce a business case and MVP roadmap. Define an approved-input value equation, sensitivity ranges, acquisition/integration costs, decision owners, Phase II milestones, DLA Component alignment options, and a transition roadmap from bounded POC to pilot-scale MVP.

Objective 6 - Demonstrate reproducibility and responsible use. Bind every reported result to a manifest, source and configuration hashes, software version, environment record, scorecard, and limitations statement. Map controls to NIST AI RMF functions and DoD Responsible AI practices without representing the POC as compliant, certified, or authorized for controlled data.

2.2 Acceptance gates

Objective	Phase I acceptance evidence
Use-case ranking	At least three use cases scored with the frozen rubric; selection rationale, dissent/uncertainty, and Component feedback recorded.
Data plan	100% of inputs labeled by source class and permitted use; minimum event schema, synthetic generator, quality rules, access assumptions, and unresolved access risks documented.
Baseline twin	Approved aggregate replay metrics evaluated against a tolerance frozen before policy selection; mismatches and unsupported dimensions retained.
Intervention evaluation	Current-state/fixed-role and cross-trained comparators run under identical scenarios; paired outcomes, intervals, negative seeds, guardrails, and abstentions reported.
Human decision support	100% of intervention cards identify assumptions, uncertainty, affected roles/processes, guardrails, rollback, and a human decision authority.
Reproducibility	Identical artifacts rerun deterministically where the model is deterministic; each scorecard resolves to source/configuration/software/output hashes.
Business case	ROI range computed only from approved inputs with sensitivity analysis; no realized-savings claim; MVP milestones, owners, dependencies, and decision gates documented.

Primary outcomes are on-time critical disposition, completion, cycle time, backlog, and bottleneck exposure. Guardrails include workload concentration, burden shift, recommendation stability, data-quality sensitivity, inappropriate use, and operator review burden. Exact service levels, replay tolerances, and decision weights will be frozen in Task 1 with the selected DLA Component or documented as approved assumptions if direct Component access is unavailable.

3. Phase I Statement of Work

3.1 Task plan

Task	Schedule	Work and outputs
1. Use-case ranking and evaluation contract	Month 1	Confirm the bounded mission question; score at least three use cases; select the POC; freeze process boundary, primary outcomes, guardrails, comparators, tolerance-setting method, data classes, and claim boundaries. Deliver use-case register, POAM, and evaluation contract.
2. Data/source/synthetic plan and controls	Months 1-2	Build the source registry, event schema, acquisition options, synthetic generator specification, missing/stale/duplicate handling, minimum-data design, export/cyber boundary, and responsible-use map. Deliver source registry, data plan, model/data cards, and risk register.
3. Baseline twin and proof of concept	Months 2-3	Implement adapters, mission-to-work graph, event replay, fixed-role and cross-trained comparators, metric engine, human-approval gate, and reproducibility manifest. Demonstrate baseline replay and record gaps without forcing a fit.
4. Scenario and intervention engine	Months 3-4	Implement surge, targeted absence, degraded-system, backlog, and human-machine teaming scenarios; add constrained routing/staffing/automation options, abstention rules, intervention cards, and rollback conditions.
5. Frozen feasibility evaluation and business case	Months 4-5	Freeze candidate policy and confirmatory protocol; execute paired holdout comparisons and perturbation/negative tests; issue scorecard and negative-result register; build approved-input ROI ranges and sensitivity analysis.
6. MVP roadmap, transition package, and final reporting	Months 5-6	Demonstrate the POC; document interfaces and deployment assumptions; produce Phase II pilot backlog, security/data dependencies, transition plan, final technical report, public final summary, and applicable patent disclosures.

3.2 Technical method

Use-case selection. Each candidate receives a 1-5 score on mission criticality, frequency and economic exposure, decision reversibility, data availability, privacy/security burden, baseline clarity, intervention controllability, and Phase II transition value. Weights and evidence requirements are frozen before scoring. Sensitivity analysis shows whether the selected use case changes under reasonable weights.

Twin representation. The POC uses an event-driven graph rather than a personnel ranking system. Process nodes describe stages and decisions; resource nodes describe roles and approved skills, not individual worth; system nodes describe availability and capacity; edges describe precedence, authorization, substitution, and information dependencies. The twin keeps an explicit difference between an observed value, a Government-approved assumption, a generated value, and a model inference.

Baseline and comparators. The current-state comparator will be defined from authorized process rules or approved assumptions. Two simple comparators remain available even when current-state data are incomplete: fixed-role FIFO and cross-trained FIFO. Any proposed routing, staffing, automation, or policy intervention is run against the same scenario seeds, demand, deadlines,

resource availability, and system state. This prevents a candidate from receiving an easier workload than its comparator.

Development/validation separation. Candidate policies may be selected only on the development partition. The chosen policy, metrics, scenario definitions, and promotion gates are then frozen. Holdout scenarios are evaluated once for the confirmatory scorecard. Any post-hoc change creates a new version and a new holdout plan; it does not overwrite the original result.

Uncertainty and negative evidence. Phase I will report paired per-scenario deltas, bootstrap or other appropriate intervals, absolute outcomes, better/tied/worse counts, and sensitivity to missing/stale data. A positive average is not sufficient if guardrails fail, losses are concentrated in critical cases, or absolute performance is operationally inadequate. Unsupported or unstable conditions trigger abstention and a stated data/capacity requirement.

Evidence receipts. Every evaluation bundle includes a source registry, protocol, configuration, software revision, environment description, metric definitions, raw scenario results, scorecard, limitations, and SHA-256 manifest. Hashes demonstrate file identity and chain-of-custody within the package; they are not represented as independent validation, immutability, certification, or proof that the underlying assumptions are true.

3.3 Preliminary generated-workflow evidence

LumenCore has implemented a public-safe generated-workflow benchmark to test the mechanics of frozen policy selection and holdout comparison. Four candidate routing policies were evaluated on 16 development scenarios. The selected policy was then held fixed across 30 disjoint validation seeds in each of five conditions and compared with fixed-role FIFO and a stronger cross-trained FIFO comparator.

Condition	Mean on-time delta vs cross-trained FIFO	Paired 95% bootstrap interval	Better / tied / worse seeds
Nominal	+0.0578	[0.0378, 0.0806]	25 / 4 / 1
Surge	+0.1156	[0.0741, 0.1611]	24 / 1 / 5
Targeted absence	+0.1175	[0.0733, 0.1638]	25 / 3 / 2
System outage	+0.1266	[0.0852, 0.1761]	28 / 0 / 2
Combined stress	+0.0302	[0.0168, 0.0435]	23 / 0 / 7

The generated result is useful because it retains the limits that a reviewer needs to see. MissionWeave was worse on on-time rate in some seeds, including 7 of 30 combined-stress seeds. Under combined stress, absolute mean on-time rate remained poor for both methods: 0.240 for cross-trained FIFO and 0.270 for MissionWeave. That condition is a capacity-breakdown region, not a solved operating mode. Mean cycle time was lower in the generated model, but no operational time, labor, readiness, or dollar effect is inferred.

The evidence bundle is located at `out/missionweave_validation/20260613T_MISSIONWEAVE_V3_DEV16_VAL30/`. The manifest file SHA-256 is `BD5FB806A6F524DE2E60D48E4D091D916F86B35B2FD73E3889667B2D8B2385DB`. Cases, workers, skills, deadlines, absences, and outages were generated. There is no DLA, personnel, timecard, workflow, or mission data; no causal identification; no validated fairness, privacy, records, or cybersecurity controls; no operator-in-the-loop production evaluation; and no 10x claim.

3.4 Deliverables

Deliverable	Planned timing
Plan of Action and Milestones / evaluation contract	Month 1 and updated monthly
Public initial project summary	At award/kickoff as directed
Monthly technical and financial status report / review	Monthly
Ranked use-case register and selection record	End of Month 1
Source registry, data acquisition/synthetic plan, schema, and control map	End of Month 2
Baseline-twin POC, source/test harness, model/data cards, and interface notes	End of Month 3
Scenario catalog, intervention cards, human-approval and abstention controls	End of Month 4
Frozen scorecard, raw paired results, negative-result register, and manifest	End of Month 5
Business case, ROI sensitivity model, and Phase II MVP roadmap	End of Month 5
Draft final report	As directed before final delivery
Final technical report, public final summary, demonstration, and applicable patent documents	End of Month 6

3.5 Risk management

Risk	Control and decision gate
No authorized DLA event data during Phase I	Use documented approved assumptions and generated data to validate software mechanics; do not issue operational or productivity claims; make data access a Phase II gate.
Model overstates an organization's causal behavior	Keep the process boundary explicit; separate observation, assumption, generation, and inference; require human interpretation; avoid a full-organizational or causal-twin claim.
Combined stress exceeds capacity	Preserve absolute outcomes, trigger abstention, identify the missing capacity or policy change, and prohibit promotion based only on relative improvement.
Missing/stale/duplicated inputs destabilize recommendations	Run perturbation tests, expose data-quality status, impose stability thresholds, and route unsupported cases to human review.
Workforce model creates inappropriate individual judgments	Model roles/process constraints, minimize personal data, prohibit protected-trait inference and employment decisions, and report burden shifts at approved aggregate levels.
ITAR/CUI enters an unapproved boundary	Use only public/generated material until the contracting office defines the data; verify JCP/DD2345, access, Technology Control Plan, and cybersecurity requirements before controlled work.

Risk	Control and decision gate
Projected CMMC Level 2 (Self) is not yet met	Do not claim assessment or compliance; maintain a pre-award gap plan and accept controlled information only after the required environment and representations are verified.
Business case uses speculative value	Require approved inputs, ranges, sensitivity analysis, and separate process metrics from mission/economic valuation; state that no savings are realized in Phase I.

3.6 Phase I price and performance

The proposed Firm-Fixed-Price total is \$100,000 for six months. The all-prime work plan assigns 640 PI hours and no subcontractor, consultant, or TABA provider. Direct and indirect labor represents \$91,200, or 91.2% of the proposed price. Remaining costs support bounded cloud/data/reproducibility services, one authorized domestic program/process-discovery trip if needed, and software/storage/evidence handling. Volume 3 controls the certified cost detail.

4. Related Work

Process mining can create action-oriented digital process representations from event data. Park and van der Aalst demonstrated an action-oriented digital twin of an organization using process-mining methods, providing a relevant foundation for connecting event evidence to operational recommendations. MissionWeave extends that direction with an explicit use-case-ranking phase, generated-scenario separation, frozen simple comparators, intervention cards, abstention, and hash-bound evidence receipts for a bounded mission process.

Research on digital twins of organizations also cautions against treating process twins as complete organizational twins. Lyytinen, Weber, Becker, and Pentland distinguish physical, business-process, and organizational twins and identify agency, conflict, symbolic interpretation, and emergence as limits to simple two-way causal representations. MissionWeave incorporates that limitation directly: it proposes a process-bounded decision instrument, not a complete causal model of DLA or its workforce.

Responsible-use controls are informed by NIST AI RMF 1.0's Govern, Map, Measure, and Manage functions and the DoD Responsible Artificial Intelligence Strategy and Implementation Pathway. Phase I translates those sources into an evaluation contract, source/assumption labeling, human approval, negative-case retention, perturbation testing, rollback, and claim boundaries. Alignment is a design objective; the proposal does not claim compliance, certification, an authorization to operate, or a completed CMMC assessment.

The existing LumenCore implementation contributes event simulation, policy comparison, deterministic configuration, tests, manifests, and reviewer-facing scorecards. Those assets reduce implementation risk, but the proposed DLA process, data agreement, security boundary, business case, and Component alignment remain Phase I work. No prior generated benchmark substitutes for Government evaluation.

5. Relationship with Future Research or Research and Development

Phase I establishes whether a bounded twin can be evaluated honestly and usefully. A successful result will include a selected DLA-relevant use case, a documented data path, a POC, evidence on where recommendations work and fail, and a business case sufficient to decide whether a Phase II pilot is warranted.

Phase II would establish a pilot-scale MVP through milestone-driven agile sprints with a relevant DLA Component. The first sprint would replace proposal assumptions with authorized process definitions and data. Later sprints would add user feedback, role-appropriate interfaces, integration

adapters, controlled-environment deployment, scenario governance, and operationally meaningful validation. Expansion to additional processes would occur only after the first process meets replay, stability, human-use, and security gates.

Illustrative Phase II milestones are: (1) Component sponsor and bounded process charter; (2) authorized data and cybersecurity boundary; (3) baseline replay accepted by process owners; (4) shadow-mode recommendations with no automated action; (5) operator evaluation and red-team review; (6) limited pilot with rollback; and (7) transition decision based on measured mission value and total cost.

Phase III would adapt the validated architecture to production orders and broader defense or commercial workflows without SBIR funding dependency. Research questions include multi-process coupling, change detection, causal identification where experimental conditions permit, privacy-preserving aggregation, robust intervention under distribution shift, and human-machine teaming governance. Each extension would retain process-specific baselines and claims rather than infer universal performance from one pilot.

6. Commercialization Strategy

6.1 Initial Government customer and transition path

The initial target user is a DLA Component leader or process owner responsible for mission readiness, workforce/process design, digital modernization, or exercise planning. Alignment with DLA J1, J3, J6, or J7 is highly desirable and will be pursued through the program; no current sponsorship or commitment is claimed. The buyer's Phase I decision is whether the bounded POC and business case justify a Phase II pilot. The Phase II decision is whether shadow-mode and limited-pilot evidence justify deployment or additional validation.

The proposed Government offering is a configurable decision-support system consisting of process/source adapters, a bounded digital twin, scenario and intervention evaluation, human-review workflow, reproducibility receipts, and implementation support. A transition package will include interface specifications, data/security prerequisites, training needs, deployment options, maintenance responsibilities, and measurable acceptance criteria.

6.2 Commercial adjacencies

Commercial and public-sector adjacencies include industrial logistics, regulated casework, field service, maintenance planning, continuity/resilience, and other workflows where delay, backlog, rework, or skill bottlenecks have measurable consequences. The same product boundary applies: configure one decision process, replay an accepted baseline, compare interventions, retain negative evidence, and expand only after a process-specific gate passes.

No customer, revenue, procurement commitment, realized savings, or independent field validation is claimed. Phase I customer discovery will test the problem, buyer, data access, deployment friction, and value equation before market claims are strengthened.

6.3 Business model and ROI method

Potential post-Phase II revenue mechanisms are fixed-price process configuration/integration, annual software licensing, and recurring validation/support. Final pricing will depend on the number of configured processes, data connectors, environment/security requirements, scenario volume, and support level.

The business case will use a bottom-up equation rather than an unsupported market multiple:

Annual net value = avoided delay/rework cost + value of additional on-time critical dispositions + avoided escalation or outage exposure - implementation and operating cost.

Each term will identify its owner, source, unit, time window, confidence range, and sensitivity. Process-time changes will not be converted to mission or dollar value without an approved mapping. The result will be a range and decision threshold, not a guaranteed return.

6.4 Commercialization milestones

Milestone	Evidence required for advancement
Phase I POC decision	Ranked use cases, data plan, frozen scorecard, negative-result register, business case, and MVP roadmap accepted for review.
Phase II pilot start	DLA Component charter, authorized data/security boundary, process owner, integration plan, and measurable acceptance contract.
Shadow-mode gate	Baseline replay accepted; recommendations stable and reviewable; no automated action; rollback and incident handling tested.
Limited-pilot gate	Approved human-use protocol, measured mission outcome, guardrails passing, cybersecurity requirements met, and transition economics updated.
Phase III/commercial release	Repeatable configuration method, support/maintenance plan, procurement path, customer evidence, and process-specific validation.

6.5 Competitive position

MissionWeave will compete on evidence discipline rather than breadth of dashboard features. Its intended advantages are bounded process selection, transparent comparator evaluation, source/synthetic separation, negative-case retention, human approval, and receipt-level reproducibility. These are proposed product characteristics supported by generated software evidence; they are not a claim of market leadership or universal model superiority.

7. Key Personnel

Robert Ashworth - Founder, Principal Investigator, and primary technical performer. Robert will lead use-case framing, architecture, implementation, evaluation, evidence packaging, business-case modeling, and reporting for 640 Phase I hours. His proposal-specific capabilities include Python automation; event and time-series simulation; heterogeneous data adapters; constrained routing; deterministic replay; SHA-256 manifests; baseline and holdout benchmark design; API/dashboard implementation; and federal-research packaging.

Relevant current artifacts include the MissionWeave benchmark harness, frozen generated-workflow validation bundle, test suite, source/evidence manifests, and proposal-specific process and claim-boundary documents described in this volume. Reviewers need not rely on external familiarity with the applicant: the preliminary evidence, exact limitations, work plan, and deliverables are contained here.

Technical education and development have been self-directed and project-based through hands-on design, implementation, testing, and documentation of software, simulation, data, and prototype systems. No degree, professional certification, security clearance, Government customer, publication, award, or granted patent is relied upon for the proposed technical evaluation.

Robert is expected to maintain the SBIR primary-employment relationship with the small business at award and during performance. Corporate-official certification controls that representation. No additional key person is proposed. If DLA identifies a required workforce, process, cybersecurity, export-control, or human-factors specialist, that need will be handled through an approved scope and cost change rather than an unsupported proposal commitment.

8. Foreign Citizens

No foreign citizen is currently planned to perform Phase I work, subject to corporate-official confirmation in DSIP. The applicant will disclose any proposed foreign citizen, country of citizenship,

immigration/work status, task, access, and location before participation and will comply with the topic's export-control requirements. No foreign person will receive export-controlled technical data through the public repository, proposal package, or unapproved development environment.

9. Facilities and Equipment

Phase I will use applicant-controlled, U.S.-located computing, version control, development tools, and evidence storage appropriate for public, generated, and other authorized unclassified inputs. The proposed work does not depend on a laboratory, classified facility, specialized manufacturing equipment, or Government-furnished hardware.

The current environment is not represented as a cleared facility, accredited CUI enclave, authorization-to-operate boundary, or completed CMMC Level 2 assessment. The topic is marked ITAR and projects CMMC Level 2 (Self). Before controlled data or covered systems are introduced, the applicant will verify JCP/DD Form 2345 eligibility, Technology Control Plan needs, access restrictions, storage/transmission controls, incident handling, and contracting-office requirements. Phase I can begin with public/generated evidence while those boundaries are resolved.

10. Subcontractors and Consultants

None are proposed. All Phase I technical work and reporting are budgeted to the small business. No letter, quote, work share, customer access, or capability of an uncommitted person or organization is used to support evaluation. Any later specialist support will require Government approval, a documented scope, and a cost treatment consistent with the award.

11. Prior, Current, or Pending Support for Similar Proposals or Awards

Effort and authoritative identifiers	Submission, award, and PI facts	Relationship to MissionWeave and cost separation
DICE. DARPA; Coherence-Bounded Peer Mesh: Sparse Task Markets and Local Inference Control for Resilient Heterogeneous AI Collectives; BAA HR001126S0010, Decentralized Artificial Intelligence through Controlled Emergence, 2026; TA1 and TA2.	Abstract submitted June 30, 2026; full proposal planned but not submitted as of July 18, 2026. No award or contract number identified. PI: Robert Ashworth, Founder and Principal Investigator.	DICE concerns decentralized heterogeneous-agent coordination, local inference control, and adversarial team evaluation. MissionWeave concerns one bounded DLA organizational process POC, business case, and transition roadmap. No MissionWeave PI hour, cloud/software charge, or deliverable is requested under DICE.
HarborSentinel. Department of the Navy; HarborSentinel: Explainable Low-Storage Pattern-of-Life Analysis for Congested Maritime Environments; DoW SBIR 2026 BAA Release 3, topic DON26BZ03-NV063.	Planned; not submitted as of July 18, 2026. No award or contract number identified. PI: Robert Ashworth, Founder and Principal Investigator.	HarborSentinel concerns maritime AIS, ADS-B, notional-radar, and composite-track anomaly review plus an SSDS transition concept. MissionWeave does not fund maritime adapters, threat-candidate logic, SSDS interfaces, or maritime-data evaluation.

Effort and authoritative identifiers	Submission, award, and PI facts	Relationship to MissionWeave and cost separation
FALCON. DARPA; Fusion of Abstract Learning and Context-Optimized Neural-methods (FALCON); DoW SBIR 2026 BAA Release 4, topic DPA26BZ04-DV016.	Direct-to-Phase-II working package; not submitted as of July 18, 2026. Local final-submission decision is NO-GO. No award or contract number identified. PI: Robert Ashworth, Founder and Principal Investigator.	FALCON concerns same-row ML-only, LLM-only, and hybrid comparisons on reserved two-domain holdouts plus hallucination, abstention, and scale measurement. MissionWeave does not fund FALCON model-fusion experiments or Direct-to-Phase-II feasibility evidence.

No prior, current, or pending award was identified as providing financial support for the MissionWeave tasks in this proposal. Shared LumenCore software is background technology; the proposed funds are for the DLA-specific process definition, adapters, evaluation, POC, business case, and transition outputs. The applicant will maintain proposal/task/day labor records and project-specific charge and deliverable ledgers. The corporate official must reconcile this table with every authoritative portal entry before certification and remove or disclose any newly identified overlap.

12. Technical Data and Software Rights Assertions

Technical data or computer software to be furnished with restrictions	Basis for assertion	Asserted rights category	Person or organization asserting restrictions
Pre-existing LumenCore computer-software components, only to the extent furnished: deterministic configuration and replay; source/provenance adapters; baseline and holdout evaluation; constrained routing and abstention; SHA-256 evidence-manifest logic; and benchmark harnesses.	Developed exclusively at private expense before any resulting award. Only components supported by dated source, funding-history, and authorship records may remain in the final assertion.	Restricted Rights in computer software, as applicable under R-DFARS 252.227-7992; SBIR/STTR Data Rights for proposal-specific software developed under a resulting award.	Robert Ashworth d/b/a LumenCore.
Pre-existing LumenCore technical data, only to the extent furnished: configuration schemas; scenario definitions; metric dictionaries; benchmark manifests; and evidence-receipt definitions.	Developed exclusively at private expense before any resulting award. Only items supported by dated source, funding-history, and authorship records may remain in the final assertion.	Limited Rights in technical data, as applicable under R-DFARS 252.227-7992; SBIR/STTR Data Rights for proposal-specific technical data developed under a resulting award.	Robert Ashworth d/b/a LumenCore.

Public standards, solicitation text, Government-furnished information, open-source dependencies, and documented Government-use interfaces are not asserted as privately developed restrictions. If

the final review cannot support an item or category, that row will be removed or corrected before certification. Completion of this table constitutes the proposal assertion under the BAA; therefore, the corporate official must review the item list, development-funding basis, category, and supporting records before submission. This table does not assert a patent, registration, Government validation, or restriction beyond the listed and supported items.

References

1. Defense Logistics Agency. DLA26BZ03-NV011, "Digital Twin of the Organization for Enhanced Mission Readiness," official DSIP topic record, 2026.
2. Department of War. DoW SBIR Program Broad Agency Announcement, 2026 Release 3, Amendment 2, 2026.
3. Defense Logistics Agency. 26.BZ Release 3 Component Instructions, Version 2, 2026.
4. Park, G., and W. M. P. van der Aalst. "Realizing A Digital Twin of An Organization Using Action-Oriented Process Mining." 2021 3rd International Conference on Process Mining, pp. 104-111. <https://doi.org/10.1109/ICPM53251.2021.9576846>.
5. Lyytinen, K., B. Weber, M. C. Becker, and B. T. Pentland. "Digital Twins of Organization: Implications for Organization Design." *Journal of Organization Design* 13, 77-93 (2024). <https://doi.org/10.1007/s41469-023-00151-z>.
6. Tabassi, E. Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST AI 100-1, 2023. <https://doi.org/10.6028/NIST.AI.100-1>.
7. U.S. Department of Defense. Responsible Artificial Intelligence Strategy and Implementation Pathway, June 2022.

Proposal Evidence Boundary

The Phase I work, costs, transition path, and controls in this volume are proposed. Preliminary quantitative results are generated-workflow software evidence only. Nothing in this proposal establishes DLA operational performance, causal workforce impact, field validation, production readiness, realized economic savings, fairness or privacy compliance, CMMC assessment, export-control certification, patent validity, independent reproduction, or a 10x improvement.